

Heterogeneity and Aggregate Fluctuations

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- Goal: develop/apply new empirical tools to generate semi-structural evidence about interaction of aggregate and distributional dynamics
⇒ VAR that includes a cross-sectional distribution.
- Prototypical questions:
 - Do distributional dynamics affect aggregate dynamics and vice versa?
 - What is the effect of aggregate shocks on cross-sectional distributions (e.g., of earnings)?
 - What are the dynamic responses to different types of distributional shocks?
- In addition:
 - Tool to assess whether current vintage of heterogeneous agent (HA) models can match joint fluctuations of macroeconomic aggregates and cross-sectional distributions.

Let's say you have some macro series (GDP, unemployment, ...) and cross-sectional data on household level labor earnings for each quarter $t = 1, \dots, T$.

Convenient and easy to use three-step approach:

- ① For each period t separately, estimate a log-spline density for the cross sectional observations X_t . This yields spline coefficients $\hat{\alpha}_t$.
- ② Then either estimate a VAR in $\hat{\alpha}_t$ and macro variables Y_t , ignoring the estimation error associated with $\hat{\alpha}_t$, or estimate a linear state-space model.
- ③ Use estimated model for $(Y_t, \hat{\alpha}_t)$ to compute IRFs etc and then re-construct cross-sectional densities.

- Framework
- Simulation Experiment: Krusell-Smith Economy
- Empirical Application: Monetary VAR
- Conclusion

The Framework

Sampling / Measurement Equation / State-space Form

- Each t : researcher observes Y_t and a sample of N iid draws x_{it} from $p_t(x)$.
(NO Panel structure: x_{it} and x_{it-1} are NOT associated with same individual!)

- Measurement equations:

$$Y_t = Y_* + \tilde{Y}_t + \text{potential measurement error}$$

$$x_{it} \sim \text{iid (over } i) p_t(x) = \frac{\exp\{\ell_*(x) + \tilde{\ell}_t(x)\}}{\int \exp\{\ell_*(x) + \tilde{\ell}_t(x)\} dx}.$$

- State-transition equations:

$$\tilde{Y}_t = B_{yy} \tilde{Y}_{t-1} + \mathbf{B}_{yl}[\tilde{\ell}_{t-1}] + u_{y,t}$$

$$\tilde{\ell}_t(x) = B_{ly}(x) \tilde{Y}_{t-1} + \mathbf{B}_{ll}[\tilde{\ell}_{t-1}](x) + u_{l,t}(x), \quad \text{e.g. } \mathbf{B}_{ll}[\tilde{\ell}_{t-1}](x) = \int B_{ll}(x, \tilde{x}) \tilde{\ell}_{t-1}(\tilde{x}) d\tilde{x}$$

- Infinite-dimensional nonlinear state-space model with states $\tilde{Y}_t$ (observed) and $\tilde{l}_t(x)$ (unobserved).

Sieve Approximation: Measurement

- We use a sieve approximation for $\ell_*(x)$ and $\ell_t(x)$:

$$\ell_t(x) \approx \ell_t^{(K)}(x) = \sum_{k=1}^K \alpha_{k,t} \zeta_k(x) = \zeta'(x) \alpha_t$$

- Define vector of sufficient statistics: $\bar{\zeta}(X_t) = \frac{1}{N} \sum_{i=1}^N \zeta(x_{it})$.
- Write (approximate) density of x_{it} , $i = 1, \dots, N$:

$$p^{(K)}(X_t | \alpha_t) = \exp \{ N \mathcal{L}^{(K)}(\alpha_t | X_t) \},$$

$$\mathcal{L}^{(K)}(\alpha_t | X_t) = \bar{\zeta}'(X_t) \alpha_t - \ln \int \exp \{ \zeta'(x) \alpha_t \} dx.$$

Sieve Approximation: State-Transition

- Operators:

$$B_{yl}(x) \approx B_{yl}^{(K)}(x) = \sum_{j=1}^K B_{yl,j} \xi_j(x) = B_{yl} \xi'(x)$$

$$B_{ll}(x, \tilde{x}) \approx B_{ll}^{(K)}(x, \tilde{x}) = \sum_{k=0}^K \sum_{j=1}^K B_{ll,kj} \zeta_k(x) \xi_j(\tilde{x}) = \zeta(x) B_{ll} \xi'(x)$$

$\vdots$

- State Transition (Approximate):

$$\tilde{Y}_t = B_{yy} \tilde{Y}_{t-1} + \mathbf{B}_{yl}^{(K)}[\tilde{\ell}_{t-1}^{(K)}] + u_{y,t}$$

$$\tilde{\ell}_t^{(K)}(x) = B_{ly}^{(K)}(x) \tilde{Y}_{t-1} + \mathbf{B}_{ll}^{(K)}[\tilde{\ell}_{t-1}^{(K)}](x) + u_{l,t}^{(K)}(x).$$

Sieve Approximation: State-Transition

We obtain

$$\begin{bmatrix} Y_t - Y_* \\ \alpha_t - \alpha_* \end{bmatrix} = \begin{bmatrix} B_{yy} & B_{y\ell} C_\alpha \\ B_{ly} & B_{ll} C_\alpha \end{bmatrix} \begin{bmatrix} Y_{t-1} - Y_* \\ \alpha_{t-1} - \alpha_* \end{bmatrix} + \begin{bmatrix} u_{y,t} \\ u_{\alpha,t} \end{bmatrix},$$

where

$$C_\alpha = \int \xi(\tilde{x}) \zeta(\tilde{x})' d\tilde{x}.$$

We collect the parameters $(\alpha_*, Y_*, B, \Sigma)$ in the vector θ .

For each t :

$$p^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta) = \int p^{(K)}(Y_t, \alpha_t | Y_{t-1}, \alpha_{t-1}, \theta) p^{(K)}(\alpha_{t-1} | Y_{1:t-1}, X_{1:t-1}, \theta) d\alpha_{t-1}$$

$$p^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta) = \int p^{(K)}(X_t | \alpha_t) p^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta) d\alpha_t$$

$$p^{(K)}(\alpha_t | Y_{1:t}, X_{1:t}, \theta) \propto p^{(K)}(X_t | \alpha_t) p^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta).$$

Approximate Filtering

- The state-transition density $p^{(K)}(Y_t, \alpha_t | Y_{t-1}, \alpha_{t-1}, \theta)$ is Normal if u_t is Normal.
- Approximate $p^{(K)}(X_t | \alpha_t)$ by second-order Taylor series expansion of $\mathcal{L}^{(K)}(\alpha_t | X_t)$ around

$$\hat{\alpha}_t = \operatorname{argmax}_{\alpha_t} \mathcal{L}^{(K)}(\alpha_t | X_t).$$

- Then

$$p^{(K)}(X_t | \alpha_t) = \exp \left\{ N \mathcal{L}^{(K)}(\hat{\alpha}_t | X_t) - \frac{N}{2} (\alpha_t - \hat{\alpha}_t)' \hat{V}_t^{-1} (\alpha_t - \hat{\alpha}_t) + N \mathcal{R} \right\}.$$

- Ignoring the remainder:

$$\hat{\alpha}_t(X_t) = \alpha_t + N^{-1/2} \eta_t, \quad \eta_t \sim N(0, \hat{V}_t).$$

⇒ Linear Gaussian state-space model, Kalman filtering.

Convenient and easy to use three-step approach:

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- ② Then either estimate a VAR in $\hat{\alpha}_t$ and macro variables Y_t , ignoring the estimation error associated with $\hat{\alpha}_t$, or estimate a linear state-space model.
- ③ Use estimated model for $(Y_t, \hat{\alpha}_t)$ to compute IRFs etc and then re-construct cross-sectional densities.

Implementation Details

- Basis functions: **cubic splines**.
- Allow for **top coded data**.
- Remove linear dependencies among $\hat{\alpha}_t$.
- **Seasonal adjustment**.
- State transition is **Bayesian VAR in multivariate regression form**:

$$W = Z\Phi + U \quad \Sigma \sim IW(\underline{\nu}, \underline{S}), \quad \phi|\lambda \sim N(\underline{\mu}_\phi, \underline{P}_\phi^{-1}(\lambda))$$

λ is a vector of hyperparameters that control the precision of the prior distribution.

- Recovering cross-sectional densities:

$$p^{(K)}(x|\alpha_t) = \frac{\exp\{\zeta'(x)\alpha_t\}}{\int \exp\{\zeta'(\tilde{x})\alpha_t\} d\tilde{x}}.$$

- Bayesian marginal data density (MDD)

$$p^{(K)}(Y_{1:T}, X_{1:T} | \lambda) = \int \left(\prod_{t=1}^T p^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta) \right) p(\theta | \lambda) d\theta.$$

- Laplace Approximation:

$$\begin{aligned} & p^{(K)}(Y_t, X_t | Y_{1:t-1}, X_{1:t-1}, \theta) \\ &= \int p^{(K)}(X_t | \alpha_t) p^{(K)}(Y_t, \alpha_t | Y_{1:t-1}, X_{1:t-1}, \theta) d\alpha_t \\ &\approx \exp \{ N \mathcal{L}^{(K)}(\hat{\alpha}_t | X_t) \} |\hat{V}_t / N|^{1/2} \times (2\pi)^{K/2} p^{(K)}(Y_t, \hat{\alpha}_t | Y_{1:t-1}, X_{1:t-1}, \theta). \end{aligned}$$

- Bayesian model selection to choose K that maximizes

$$p^{(K)}(Y_{1:T}, X_{1:T}|\lambda) \approx (2\pi)^{KT/2} \left(\prod_{t=1}^T \exp \{ N\mathcal{L}^{(K)}(\hat{\alpha}_t|X_t) \} |\hat{V}_t/N|^{1/2} \right) \\ \times \int \left(\prod_{t=1}^T p^{(K)}(Y_t, \hat{\alpha}_t|Y_{t-1}, \alpha_{t-1} = \hat{\alpha}_{t-1}, \theta) \right) p(\theta|\lambda) d\theta.$$

Simulation Experiment: Data from a Krusell-Smith Economy

Krusell-Smith (KS) Economy

- **Heterogeneous households** $j \in [0, 1]$, CRRA utility over consumption, asset demand $a_t(j)$ is function of aggregate return r_t , **borrowing constraint** $a_t(j) \geq 0$.
- **Households supply** $\epsilon_t(j)$ **efficiency units of labor (inelastically)**, $\epsilon_t(j) \in \{0, 1\}$, two-state Markov process:
 - employed: $\epsilon_t(j) = 1$, after labor-tax earnings $(1 - \tau) \cdot w_t$
 - unemployed: $b \cdot w_t$
 - aggregate labor: $L = \int \epsilon_t(j) dj$ is constant.
- **Representative firm**: $Y_t = e^{z_t} K_t^\alpha L^{1-\alpha}$, $r_t = \alpha Y_t / K_t$. Technology: $z_t = \rho_z z_{t-1} + \sigma_z \epsilon_t$.
- **Aggregate state** s : z , distribution μ over (ϵ, a) .
- **Market clearing**: $Y_t = \int c_t(j) dj$, $K_t = \int a_t(j) dj$.

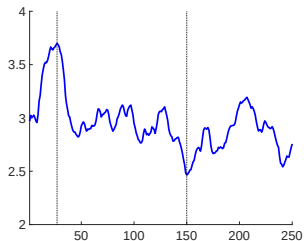
- Winberry (2017)
- Approximate the stochastic steady state, which involves:
 - approximating the cross-sectional density

$$p_{t,\epsilon}^{(K)}(a) = \exp \left\{ \alpha_{t,\epsilon,0} + \alpha_{t,\epsilon,1}(a - m_{t,\epsilon,1}) + \sum_{k=2}^K \alpha_{t,\epsilon,k} \left[(a - m_{t,\epsilon,1})^k - m_{t,\epsilon,k} \right] \right\}.$$

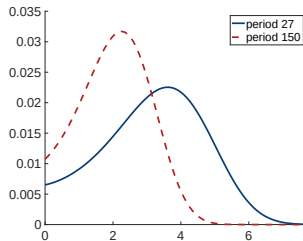
- First-order perturbation approach to approximate the aggregate dynamics of the system.
- State variables: aggregate productivity, mass of agents with zero assets, $\alpha_{t,k}$ that characterizes asset distribution.
- Simulate data and estimate functional state-space model.

KS Simulated Data

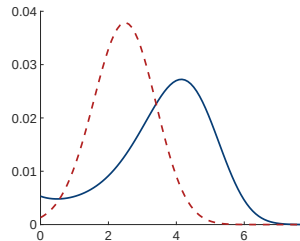
Capital Stock
Time Series



Asset Distribution
Unemployed



Asset Distribution
Employed



Knot Placement

K	Percentiles								
	1	2.5	5	10	25	50	75	90	95
4					X	X	X		
6				X	X	X	X	X	
8			X	X	X	X	X	X	X
10	X	X	X	X	X	X	X	X	X

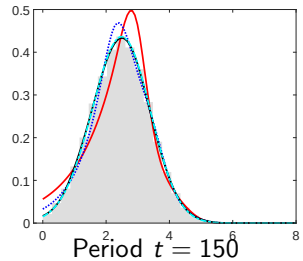
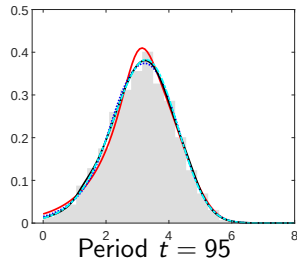
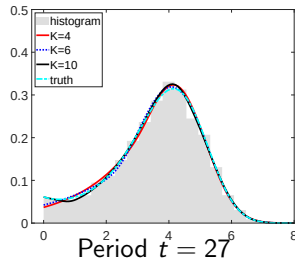
Do Distributional Dynamics Affect Aggregate Dynamics?

- State-transition VAR:

$$\begin{bmatrix} \tilde{Y}_t \\ \tilde{\alpha}_t \end{bmatrix} = \begin{bmatrix} \Phi_{yy} & \Phi_{yl} = 0? \\ \Phi_{ly} = 0? & \Phi_{ll} \end{bmatrix} \begin{bmatrix} \tilde{Y}_{t-1} \\ \tilde{\alpha}_{t-1} \end{bmatrix} + \begin{bmatrix} u_{y,t} \\ u_{\alpha,t} \end{bmatrix},$$

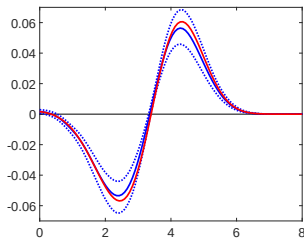
- Prior is centered at zero. $\phi|\lambda \sim N(\underline{\mu}_\phi, \underline{P}_\phi^{-1}(\lambda))$
- Use precision hyperparameters to control degree of shrinkage by maximizing $p(Y|\lambda)$:
 - $K = 10$
 - λ_1 overall precision: $\hat{\lambda}_1 = 1.2$
 - λ_2 controls relative precision for Φ_{yl} block: $\hat{\lambda}_2 = 403 \implies \Phi_{yl} = 0$
 - λ_3 controls relative precision for Φ_{ly} block: $\hat{\lambda}_3 = 1.2 \implies \Phi_{ly} \neq 0$
- Conclusion:** Shrinkage is strongest for Φ_{yl} block which is zero in DGP.

Cross-sectional Fit ($N = 10,000$, $T = 250$, $K = 10$ Selected)



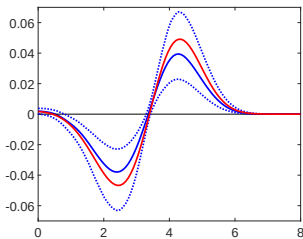
Impulse Responses to a 3-Standard-Deviation Technology Shock

Density Differential
 $h = 5$



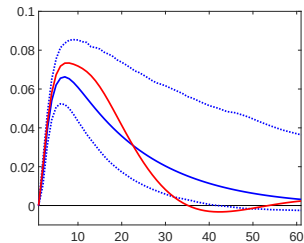
Asset holding x

Density Differential
 $h = 15$



Asset holding x

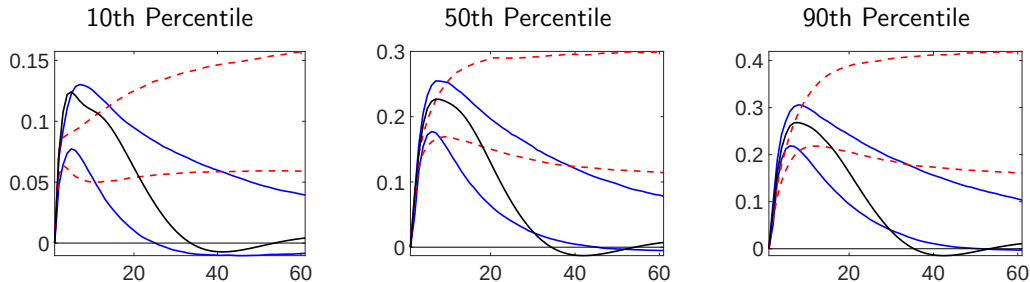
Fraction of Individuals
With Assets > 4



Time horizon h

Functional vs. Alternative VAR: Percentile Responses to a TFP Shock

- Compare Functional VAR with an alternative VAR with aggregate data + percentiles (10, 20, 50, 80, 90th)
- The alternative VAR fails to capture the underlying truth.



Notes: "True" responses from KS model in solid black; 80% band computed from the functional VAR in solid blue; 80% bands from Alternative VAR in dashed red.

Empirical Application

A Monetary VAR

Empirical Analysis with a Monetary VAR: Questions

- ① Do Distributional Dynamics Affect Aggregate Dynamics?
- ② What is the effect of a monetary policy shock on the cross-sectional distribution of earnings (or consumption)?

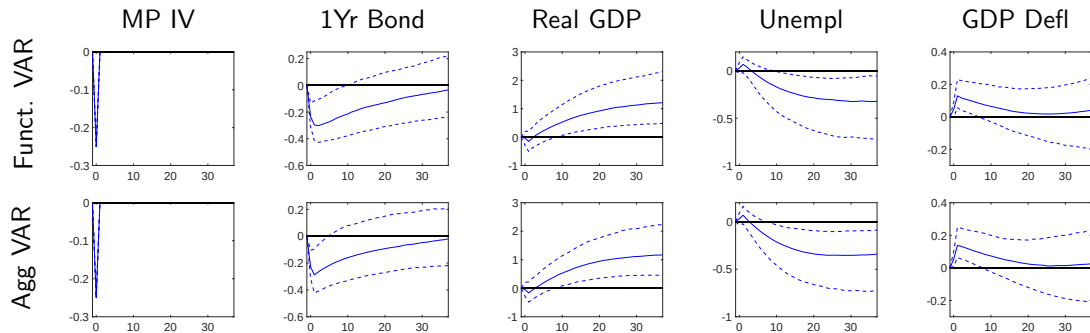
Empirical Application: A Monetary VAR

- **Starting point:** VARs by Jarocinski and Karadi (2020, AEJ: Macro).
- Identify two structural shocks transmitted through the central bank announcements.
 - (1) **Monetary policy shock** based on the **negative comovement** of interest rates and stock prices.
 - (2) **Central bank's information shock** based on the **positive comovement** of interest rates and stock prices.
- **VAR aggregate variables:**
 - High-frequency instrument(s): fed funds futures surprises, S&P500 surprises
 - Macro variables: 1-year gov. bond yield, S&P500, real GDP, GDP deflator, excess bond premium, unemployment rate.

Empirical Application: A Monetary VAR

- Repeated cross sections:
 - Nominal earnings (CPS), standardized by $2/3$ (approx labor share) of per capita GDP.
 - $K=10$ spline coefficients, representing the continuous part.
 - Normalization applied to make point mass at zero equal to unemployment rate.
- Sample periods: 1990:M2 to 2016:M12.

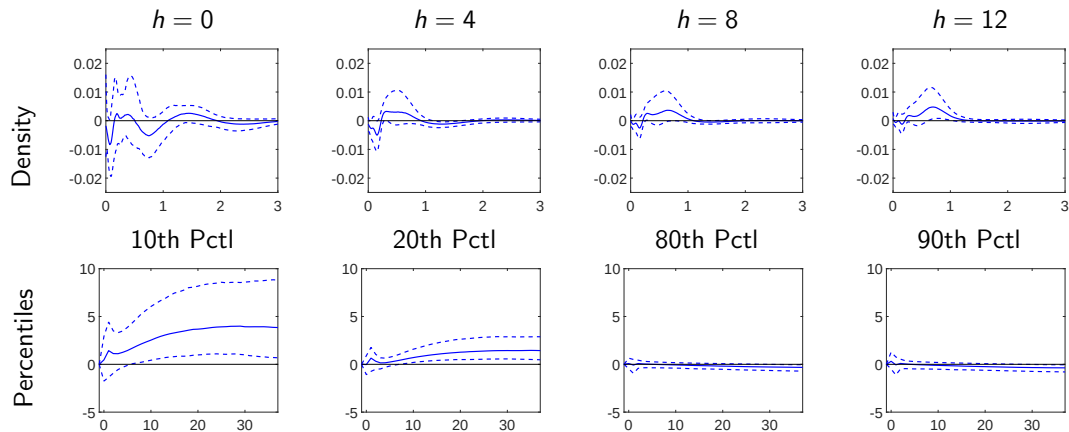
Responses of Aggregate Variables to JK Monetary Policy Shock



Impulse: Responses to a 25bp monetary policy shock. System is in steady state at $h = -1$, shock occurs at $h = 0$.

Finding: Responses of Agg Vars. to JK-MP shock are essentially identical.

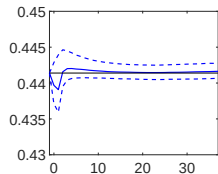
Response of Earnings to JK Monetary Policy Shock



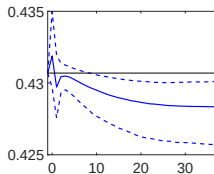
Finding: Largest effect in the left tail of distribution.

Responses of Inequality Measures to JK Monetary Policy Shock

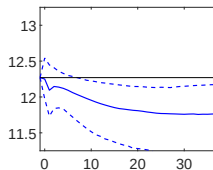
Fraction Earning Less Than GDP Per Capita



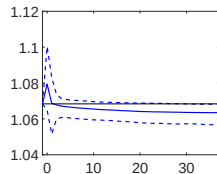
Gini Coefficient



90%-10% Ratio



Cross-sectional Standard Deviation



Finding: While the short-run effect is ambiguous, the inequality measures generally fall in the medium run.

- Analysis focused on **indirect effects of MP on inequality** (Kaplan, Moll, and Violante, 2018):

Mon. policy($\uparrow$) $\longrightarrow$ real R($\downarrow$) $\longrightarrow$ Demand($\uparrow$) $\longrightarrow$ Labor, Income($\uparrow$).

- Consistent with HA model with indivisible labor supply as in Chang and Kim (2006) and Ma (2021) in HANK:

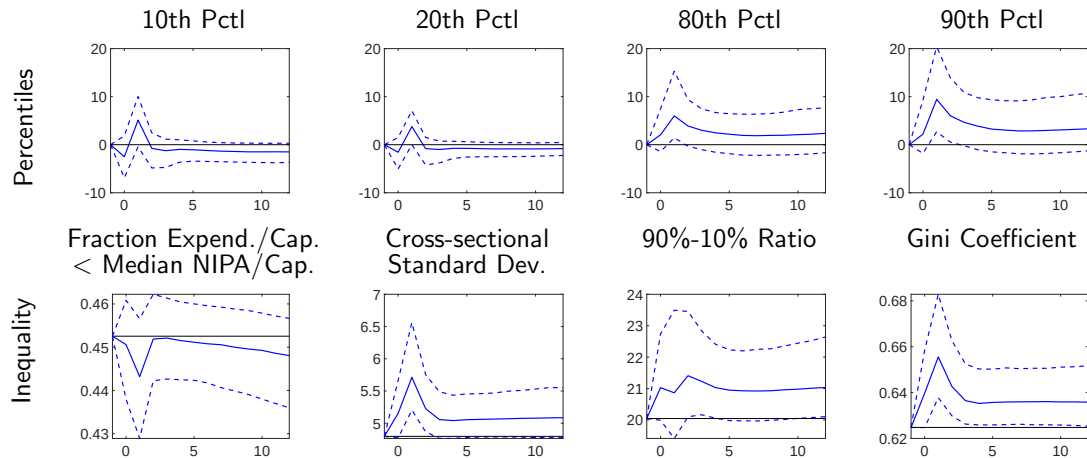
During booms wages rise and exceed the reservation wage of low-skilled workers.

- **Reduction in earnings inequality is mainly driven by fall in unemployment.** Inequality measures computed from cont. part of distribution move less.
- Compare Gini drop to literature: **we report .001 to .002**, Coibion, Gorodnichenko, Kueng, and Silvia (2017) obtain .0006, Furceri, Loungani, and Zdzienicka (2018) report .005.

Empirical Application: A Monetary VAR

- Repeated cross sections:
 - Consumer Expenditure Survey (BLS)
 - Quarterly per capita consumption expenditures are computed by summing within-quarter monthly expenditures.
 - Standardized so that 1 means approximately consuming at the level of NIPA per capita.
 - K=10 spline coefficients, representing the continuous part.
- Sample periods: 1990:Q2 to 2016:Q4.

Results with CEX Consumption Data, JK Shock



Finding: Unlike earnings inequality, consumption inequality generally rises in response to expansionary MP shock.

- In response to an expansionary monetary policy shock, consumption inequality increases. Some potential explanations:
- Rich households typically also have a significant amount of unearned income, which may be directly affected by the monetary policy intervention.
- Even wealthy households who have a relatively high level of consumption may be close to their borrowing constraint (wealthy hand-to-mouth consumers) and have a relatively high MPC.
- High consumption/income households' decisions are more interest rate sensitive.

- We develop/apply new empirical tools to generate semi-structural evidence about interaction of aggregate and distributional dynamics.
- Functional autoregressive model that accounts for the fact that the functions, i.e., cross-sectional (log) densities are themselves unobserved.
- Many applications... we considered the interaction between macroeconomic aggregates and distribution of earnings.
- Can be used to evaluate HA models; blue print for likelihood construction.
- Many extensions... mixed frequency observations, account for correlations in cross-sectional data.

Some References – Quantitative Research

- **Do distributional dynamics affect aggregate dynamics and vice versa?**
 - Ahn, Kaplan, Moll, Winberry, and Wolf (2017: HA Model): compare propagation of prod. shocks in models with/without heterogeneity.
 - Kaplan and Violante (2018: HA Model): propagation of prod. and monet. policy shocks in HANK, RANK, and TANK models.
 - Ottonello and Winberry (2018: HF Model): aggregate effect of monet. policy in model with heterogeneity in firms' financial positions.
- **What is the effect of an aggregate shock on cross-sectional distributions?**
 - Ahn, Kaplan, Moll, Winberry, and Wolf (2017: HA Model): factor-specific prod. shocks $\mapsto$ inequality dynamics.
 - Kaplan and Violante (2018: HA Model): monet. policy shock $\mapsto$ consumption distribution.
 - Coibion, Gorodnichenko, Kueng, and Silvia (2017: VAR with cross-sec moments): monet. policy shock $\mapsto$ earnings, income, consumption inequality.
 - Mongey and Williams (2017: HF Model): macro shocks $\mapsto$ dispersion of sales growth.
- **Effect of a change in a cross-sectional distribution on macroeconomic aggregates?**
 - Auclert and Rognlie (2018 WP: HA Model): exogenous rise in inequality $\mapsto$ macro dynamics.

Some References – Technical Research

- **Analysis of functional time series.** Function is typically treated as observed.
Monographs: Bosq (2000), Ramsey and Silverman (2005), Horvath and Kokoszka (2012).
Applications in econ: Meeks and Bowsher (2008), Chang, Kim, and Park (2016, 2018), Hu and Park (2017), Meeks and Monti (2019); also Diebold and Li (2006) and Inoue and Rossi (2017).
- **Functional analysis of pdfs.** Bayes Hilbert spaces and centered log ratio transformation: E.g., van den Boogaart, Egozcue, Pawlowsky-Glahn (2014).
- **Sieve approximations in econometrics.** Log-spline densities: Kooperberg and Stone (1990); survey: Chen (2007, HB Econometrics).
- **Solution of HA macro models.** Cross-sectional distribution of skills and assets is a state variable that affects agents' decision. Representation by moments: Krussel and Smith (1998), Algan, Allais, and Den Haan (2008), Winberry (2017); discretization: Reiter (2010); wavelets: Childers (2018).